

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5Total Marks:25

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

1.Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

Comparative Analysis of Star Topology and Bus Topology

Feature	Star Topology	Bus Topology
Cost	More expensive because it requires a central device such as a switch or hub and separate cables for each node.	Less expensive because it uses a single backbone cable to connect all devices.
Installation	Requires more cabling and configuration, making installation comparatively complex.	Easy to install because all devices share a common backbone cable.
Reliability	More reliable because failure of one node or cable usually does not affect the other nodes.	Less reliable because failure of the main backbone cable can bring down the entire network.
Performance	Provides better performance because the central switch manages data transmission and reduces collisions.	Performance decreases as more devices are added because all devices share the same communication medium.
Failure	Failure of a single node affects only that particular node. However, failure of the central switch can affect the entire network.	Failure of the backbone cable can cause the entire network to stop working.
Scalability	Easy to add new devices by connecting them to the central switch.	Adding more devices can increase traffic and reduce network performance.

2.Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

Comparative Analysis of Star Topology and Bus Topology

Parameter	Mesh Topology	Star Topology
Fault Tolerance	Very high because multiple paths are available between nodes.	Moderate because the central hub/switch is a single point of failure.
Reliability	Highly reliable and suitable for critical networks.	Reliable under normal conditions but depends on the central device.
Failure of a Link	Failure of one link usually does not affect communication because alternate paths exist.	Failure of a cable affects only the connected node.
Failure of Central Device	No central device, so there is no single point of failure.	Failure of the hub/switch can bring down the entire network.
Cost	High cost because many cables and network interfaces are required.	Lower cost because each device requires only one connection to the central device.
Installation	Difficult and time-consuming due to multiple connections.	Easy and simple to install.
Maintenance	Difficult because there are many connections to monitor.	Easy because problems can be identified through the central hub/switch.
Performance	Provides good performance and alternative routes, especially in critical networks.	Provides good performance, but heavy traffic through the central device can cause congestion.

3.Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

Comparative Analysis: Bus Topology vs Mesh Topology

Parameter	Bus Topology	Mesh Topology
Installation	Simple and easy to install because it uses a single backbone cable.	Complex to install because each node requires multiple connections.
Cost	Low cost due to fewer cables and devices.	High cost due to large number of cables and ports.
Complexity	Simple network structure with minimal connections.	Highly complex because nodes are interconnected.
Number of Cables	Requires a single main backbone cable.	Requires many cables to connect nodes.
Time Required for Setup	Quick to install and configure.	Time-consuming due to numerous connections.
Maintenance	Relatively easy for small networks.	Difficult because many links must be monitored.
Scalability	Suitable mainly for small networks.	Can support larger networks but becomes increasingly complex and costly.
Fault Tolerance	Low; backbone failure can stop communication.	Very high; failure of one link does not necessarily disrupt communication.

4.Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

Comparative Analysis: Hybrid Topology vs Star Topology

Parameter	Hybrid Topology	Star Topology
Scalability	Very high; new network segments can be added flexibly.	Moderate; expansion depends on the capacity of the central switch/hub.
Network Expansion	Easy to expand by adding new topology segments.	Limited by the available ports on the central device.
Flexibility	Highly flexible because different topologies can be combined.	Less flexible because all devices depend on the central device.
Cost	Generally high due to different devices and network structures.	Lower compared to hybrid topology.
Complexity	More complex because multiple topologies are integrated.	Simple and easy to understand.
Installation	Complex and requires careful planning.	Simple and easy to install.
Reliability	High; failure in one segment may not affect other segments.	Moderate; failure of the central switch/hub can affect the whole network.
Fault Tolerance	High depending on the topologies used.	Moderate.

5.In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

Comparative Analysis: Star vs Mesh vs Bus vs Hybrid Topology — Maintenance

Parameter	Star Topology	Mesh Topology	Bus Topology	Hybrid Topology
1. Maintenance	Easy	Difficult	Difficult	Moderate
2. Fault Identification	Easy and quick	Difficult	Difficult	Moderate
3. Troubleshooting	Simple and quick	Complex and time-consuming	Difficult	Moderately complex
4. Failure Impact	Usually affects only one node	Usually limited due to alternate paths	Backbone failure can affect the entire network	Depends on the affected segment
5. Maintenance Cost	Low	High	Low–Moderate	Moderate–High
6. Network Structure	Simple and centralized	Complex and interconnected	Simple with backbone cable	Complex with multiple topologies
7. Overall Maintenance Complexity	Low	Very High	High	Moderate